Reflexive Reality and Self-Defining Physical Law

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Abstract

Modern physics treats the laws of nature as meta-descriptions written "outside" the universe. If the universe is complete and self-contained, its governing principles must arise *from within*: the laws, the state, and the act of evolution are three aspects of a single reflexive process. This paper formalises the concept of *reflexive reality*, classifies known physical theories by their degree of reflexivity, and illustrates with concrete examples how self-defining dynamics manifest. The central philosophical argument is that a finite, knowable, self-contained universe must obey reflexive laws [**I**]. This argument has been formalised and machine-checked in the companion NEMS programme: the Theorem of Foundational Finality (NEMS 23, `NemS.foundational_finality`, Zenodo 10.5281/zenodo.19429761) [**T**] and the Reflexive Closure Theorem (NEMS 56, `ReflexiveClosure`, Zenodo 10.5281/zenodo.19429835) [**T**] provide the machine-checked backing. This paper presents the physical intuition and conceptual framework; readers seeking formal proofs are directed to the NEMS companion papers [1].

1. Introduction: Why Reflexivity Matters

Modern physics typically treats the universe as a system evolving under externally defined equations. The laws are written "outside" the universe, as meta-descriptions that govern its behaviour. Yet if the universe is complete and there is no "outside," then its governing principles must arise *from within*. The laws, the state, and the act of evolution must be three aspects of a single, self-contained process.

This requirement leads to the concept of **reflexivity**: the universe must include within itself the full definition of the laws that determine it. A **reflexive reality** is one that contains and executes its own generative rules; its physical laws are not externally imposed but are self-defining. Formally, this corresponds to the fixed-point condition

$$S = L(S),$$

where S denotes the total state of the universe and $L(S)$ is the meta-description or language that defines its own evolution. The universe is therefore its own equation of motion.

The purpose of this article is to formalise what reflexivity means in physics, to classify degrees of reflexivity among known laws, and to illustrate with concrete examples—both classical and from the Reflexive Reality (RR) programme—how self-defining dynamics manifest in physical systems.

Position in the series. This paper presents the philosophical framework motivating the formal results developed in the companion NEMS programme (Papers 01–92, Zenodo [1]). Formal proofs of the claims outlined here are in NEMS Papers 02, 23, 56, and 76 [2, 3, 4, 5].

2. Definition of Reflexivity

Definition 1 (Reflexive System). [**I**] A system S is *reflexive* if its description contains a representation of the generative rules that define its evolution. Formally,

$$S = L(S),$$

meaning that the language, rule set, or law that defines S is also instantiated within S. Such systems achieve **Perfect Self-Containment (PSC)**: they require no external meta-law for their operation.

Definition 2 (Reflexive Law). [**B**: `NemS.classification_theorem`, Zenodo 10.5281/zenodo.19429715 [2]] A physical law $\mathcal{L}$ acting on a state Ψ is *reflexive* if (a) the operator $\mathcal{L}$ depends functionally on the same state it evolves,

$$\frac{d\Psi}{dt} = \mathcal{L}[\Psi], \qquad \mathcal{L} = \mathcal{F}[\Psi],$$

(b) the structural evolution is self-consistent under this dependence, and (c) the defining parameters of $\mathcal{L}$ (coupling constants, boundary conditions) are themselves determined by the system's dynamics, admitting no external free-parameter completion (the "no free bits" condition formalised in NEMS Paper 02 [2]).

A law of this form is self-modifying; its structure changes in response to the state it governs, maintaining global coherence.

Non-reflexive laws are those where the operator is fixed externally—such as a classical Hamiltonian with constant parameters independent of the system's internal configuration.

3. Reflexivity in Known Physical Theories

Example 1 (Newtonian Mechanics). [**I**] Newton's laws assume an external Euclidean space, fixed constants, and absolute time. Neither the geometry nor the constants depend on the system's internal state. Thus,

$$\ddot{x} = -\nabla V(x), \quad V(x) \text{ fixed},$$

is *non-reflexive*: the law does not contain its own boundary conditions or the metric it presupposes.

Example 2 (General Relativity). [**I**] Einstein's field equations

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$$

are *weakly reflexive.* Geometry ($G_{\mu\nu}$) and matter ($T_{\mu\nu}$) co-determine each other: the distribution of energy–momentum shapes spacetime curvature, which in turn constrains the motion of matter. However, constants such as G and the metric signature are externally fixed, so the theory is not fully reflexive.

Example 3 (Quantum Mechanics). [**I**] The Schrödinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$$

is not reflexive because the Hamiltonian $\hat{H}$ is externally defined. In a reflexive reformulation,

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}[\psi]\psi,$$

the Hamiltonian depends on the state's informational or geometric curvature, making the evolution self-adjusting. Wavefunction collapse then emerges as a reflexive dissonance-minimisation event rather than an external measurement postulate.

Example 4 (Thermodynamics). [**I**] Classical thermodynamics relies on externally defined ensembles and boundary conditions. However, when entropy and temperature are defined through information exchange internal to the system, such as in self-organising nonequilibrium systems, the law becomes partially reflexive: the gradients that drive evolution are produced by the system's own state history.

Example 5 (The SDS and Principle of Coherence). [**B**] In the Self-Defining System (SDS) framework, dynamics arise from minimisation of the Ontological Dissonance functional:

$$D = w_1 D_{\text{inc}} + w_2 D_{\text{comp}} + w_3 D_{\text{temp}} + w_4 D_{\text{clos}},$$

and the evolution of the universe is given by

$$\frac{d\Psi}{dt} = -\frac{\delta D[\Psi]}{\delta \Psi}.$$

Here the generator of motion is a functional of the state itself. The universe continuously measures its own inconsistency and adjusts to minimise it. This gradient-flow equation approximates the system near equilibrium; the full dynamics at choice points require Transputation for unitarity-preserving selection [**T**: `Transputation.Theorems.ForcedAdjudication`, Zenodo 10.5281/zenodo.19429882 [5]]. The resulting law is fully reflexive: both the metric of evaluation and the process of correction are contained within the system.

4. Degrees of Reflexivity

The following table classifies physical theories by their degree of reflexivity [**I**: informal taxonomy; formal version in NEMS 02 [2], Theorem `NemS.classification_theorem`].

Framework or Law	External Dependence	Reflexivity Level	NEMS Class
Newtonian Mechanics	Fixed space and constants	Non-reflexive	Class I
Quantum Mechanics (standard)	External Hamiltonian	Non-reflexive	Class I
Thermodynamics (classical)	External reservoir	Weakly reflexive	Class I
General Relativity	Mutual geometry–matter coupling	Weakly reflexive	Class I–IIa
Quantum Mechanics (state-dependent $\hat{H}$)	State-dependent operator	Strongly reflexive	Class IIa
Self-Defining System / MFRR	Fully self-referential dynamics	Fully reflexive	Class IIb

Note on NEMS classes. NEMS Paper 02 proves that all autonomous self-modelling systems fall into three classes: Class I (externally determined), Class IIa (restricted internal selection), Class IIb (non-effective internal adjudication; Transputation forced by NEMS 08/09). The assignment above is informal; the formal class assignment for Class IIb systems requires checking the PSC axioms A0–A5 of NEMS 02 [2].

5. Worked Examples of Reflexive Laws

Worked Example 1: The Coherence–Information Law. [**B**] In the MFRR framework, which builds on SDS, gravitational and informational dynamics are unified:

$$G_{\mu\nu} = 8\pi G\,(T_{\mu\nu} + \mathcal{C}_{\mu\nu}),$$

where $\mathcal{C}_{\mu\nu}$ is the *coherence stress–energy tensor* derived from the global information field $\Phi[\Psi]$. Because $\mathcal{C}_{\mu\nu}$ depends on the state of informational coherence of the universe, the curvature of spacetime responds not only to energy and momentum but to the system's own informational structure. The law is reflexive: geometry, energy, and information are dynamically co-determining.

Worked Example 2: The Reflexive Landauer Bound. [**B**] The minimal energetic cost of maintaining consistency between a system's state and its self-description is

$$\Delta E \geq k_B T \ln(2)\,\Delta\Psi_\Omega,$$

where $\Delta\Psi_\Omega$ measures the change in global coherence. The constant k_B and temperature T enter as emergent averages of the same process; the law expresses the cost of self-maintenance in any reflexive computation.

Worked Example 3: Choice–Curvature Correspondence. [**I**] Each act of information adjudication (Transputation) at a local choice point introduces a curvature perturbation:

$$\delta R \propto \nabla_i \nabla_j \Delta\Phi.$$

Here, a cognitive or informational decision is inseparable from a geometric deformation. This provides a physical bridge between consciousness and spacetime structure—both are modes of coherence correction within the same reflexive field.

Worked Example 4: Reflexive Quantum Harmonic Oscillator. [**B**] In a reflexive extension of the harmonic oscillator,

$$i\hbar\frac{\partial\psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\nabla^2 + \frac{1}{2}m\omega^2(\Phi[\psi])x^2\right]\psi,$$

the frequency ω depends on the global coherence $\Phi[\psi] = \int |\psi|^2 \ln |\psi|^2 \, dx$. When $\Phi[\psi]$ decreases (greater coherence), the potential relaxes; the oscillator "tunes itself" to maintain constant dissonance. This is a well-defined nonlinear Schrödinger equation; a full worked computation of its stationary states and energy-spectrum corrections is an open technical target.

6. Reflexive Closure Proposition

Proposition (Informal; Reflexive Closure). [**B**: formal version is **T** NEMS 23+56 [3, 4]] A finite, self-contained, and knowable universe must obey reflexive laws. If a system's governing equations are not reflexive, then the definitions of its constants or boundary conditions require an external meta-law, contradicting self-containment.

Grounding sketch. Let S be the universe and L its governing law. If $S \neq L(S)$, there exists a higher-order law L' defining L. If L' is within S, the regress continues indefinitely; if L' is external, S is not closed. The NEMS programme replaces the informal "regress" step with a machine-checked trichotomy: any purported external meta-explanation of a PSC universe must either violate PSC, be physically redundant, or be isomorphic to the universe itself (Theorem of Foundational Finality, NEMS Paper 23 [**T**], `NemS.foundational_finality`, Zenodo 10.5281/zenodo.19429761 [3]). NEMS Paper 56 (Reflexive Closure Theorem [**T**], `ReflexiveClosure`, Zenodo 10.5281/zenodo.19429835 [4]) further proves that a reflexive system may close but cannot coincide with its own complete semantic image—self-exhaustion is impossible. Together these results formalise and strengthen the intuition of the sketch above.

7. Consequences and Empirical Manifestations

Reflexivity predicts and explains several observed phenomena:

- **Stability of constants:** [**I**] Physical constants correspond to attractor points in the dissonance landscape defined by $D[\Psi]$.
- **Quantum–classical transition:** [**I**] Loss of local reflexivity (weakening of global coherence coupling) produces classical decoherence.
- **Information–gravity coupling:** [**B**] Curvature responds to informational structure, as explored computationally in companion work on SDS field experiments [1].
- **Wavefunction collapse:** [**B**] Transputation selects globally coherent branches, resolving superpositions without stochasticity [5].

8. Summary: Physics as Self-Definition

A **self-defining physics** is one in which:

1. The laws of nature are themselves emergent solutions of the universe's consistency conditions.
2. Constants, symmetries, and "initial conditions" are equilibrium values of a self-correcting reflexive process.
3. The evolution of the universe is an ongoing proof of its own coherence:

$$\frac{d}{dt}(\text{Consistency of } S) = 0.$$

4. Observers and cognition are not anomalies but the physical means by which the universe maintains that consistency.

In a Reflexive Reality, *law, state, and observation form a single closed loop of self-definition.* The cosmos is not executed by an external runner; it is a living proof of its own theorem—a universe that computes, perceives, and refines itself through the continual act of maintaining coherence.

Companion papers. The MFRR framework that grounds this reflexive-law analysis is surveyed for physics audiences in [6] (P20), covering the T8 holographic closure test, the IPT threshold, and the RCC theorem, together with Standard Model parameter derivations and Lean-certified proofs of the core closure theorems. The extended formal treatment—full proofs, 297 Python modules, and 57,337+ experimental runs—is in the foundational MFRR monograph [7] (P13).

Appendix: Formal Backing in nems-lean

The following theorems in the nems-lean library (Zenodo 10.5281/zenodo.19429710, GitHub: `https://github.com/novaspivack/nems-lean`) provide machine-checked formal backing for the philosophical claims of this paper. All are proved with zero `sorry` and zero custom axioms; build command: `lake update && lake build` from the nems-lean root.

`NemS.classification_theorem` **(NEMS Paper 02, Zenodo 10.5281/zenodo.19429715 [2]).** *Statement:* Under PSC axioms A0–A5 (no external selection, no free bits, semantic closure), all autonomous self-modelling systems fall into exactly three classes: Class I (externally determined), Class IIa (restricted internal selection), Class IIb (non-effective internal adjudication). This formalises Definition 2's "no free bits" clause (c) and the NEMS class column of the §4 taxonomy table.

`NemS.foundational_finality` **(NEMS Paper 23, Zenodo 10.5281/zenodo.19429761 [3]).** *Statement:* Any attempt to explain a PSC universe from the "outside" must either violate self-containment, be physically redundant (adding no new record-truth), or be isomorphic to the universe itself. This is the machine-checked formalisation of the Reflexive Closure Proposition in §6, replacing the informal "infinite regress" argument with a trichotomy.

`ReflexiveClosure` **(NEMS Paper 56, Zenodo 10.5281/zenodo.19429835 [4]).** *Statement:* A reflexive system may achieve closure but cannot coincide with its complete semantic image: self-exhaustion is impossible. This rules out the scenario in which a reflexive universe could "fully know itself" in a way that breaks the fixed-point structure $S = L(S)$.

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[7] Nova Spivack. Mathematical foundations of reflexive reality. Preprint (forthcoming on Zenodo); available at https://github.com/novaspivack/ugp-physics under `papers/13_MFRR_foundational_monograph/`, 2026. Full proofs and computational experiments for the MFRR programme. The 12-page journal survey (P20) provides a self-contained summary; see https://github.com/novaspivack/ugp-physics.